Controlled experiments: coverage and findings

216 recordings · 2.9 h · 6 participants

FACTOR	CONDITIONS	N REC.	OBSERVATION
Viewing distance	1, 2, 4, 5, 8, 12 m	**79**	**P1 +0.58°; P2 0.96 → 2.27°** Significant accuracy change; stable dispersion
Vertical position	Upper / lower grid	**82**	**P1 +1.75°; P3 +1.01°** Lower-row RMS-S2S higher in all 6; $p < .05$
Eccentricity	Center to 30°	**82**	**No robust peripheral loss** Accuracy stable across tested regions
Bright / dark	26 bright / 26 dark	**52**	**No robust accuracy change** Pupil size changes without accuracy loss
Indoor / outdoor	20 indoor / 20 outdoor	**40**	**Accuracy comparable** Dispersion 0.23° → 0.27° ($p = .013$)*
Smooth pursuit	8 directions · 3 speeds	**58**	**Residual spread 0.27° → 0.93°** 181 smooth pursuits; $p = .001$
Repetition	Across sessions	**34**	**No within-day offset drift** Repeated observations across days

Counts overlap. Pursuit: 58 comparison recordings (50 static + 8 pursuit); 181 counts individual smooth pursuits.
Distance effects use the analyzed subset; wider distance sweeps are included in coverage, not yet in factor tests.
* Indoor/outdoor dispersion is from the across-day comparison; matched-day accuracy was comparable.